Best acceptable settings on the Kp × α grid (overshoot ≤ 6 %, settling ≤ 10 s), ranked by PWM jitter

Rank	Setting	Kp	Ki	α	PWM jitter (% duty)	vs baseline	Overshoot (%)	Settling (s)	Speed ripple (RPM)	ripple vs baseline	neighbours acceptable	runs
base	A_baseline	2.629	1.454	1.0	0.177	—	4.2	16.44	3.16	—	—	5
1	G44_kp2.235_a0.2	2.235	1.454	0.2	0.032	−82 %	4.3	6.28	4.36	+38 %	3 of 9	1
2	G27_kp3.155_a0.15	3.155	1.454	0.15	0.034	−81 %	4.2	7.07	3.10	−2 %	4 of 6	1
3	G35_kp2.6294_a0.2	2.629	1.454	0.2	0.037	−79 %	4.3	6.47	3.22	+2 %	5 of 9	1
4	G18_kp3.813_a0.15	3.813	1.454	0.15	0.041	−77 %	4.0	7.95	3.38	+7 %	3 of 6	1
5	G17_kp3.813_a0.2	3.813	1.454	0.2	0.055	−69 %	3.7	7.68	3.12	−1 %	5 of 9	1
6	G34_kp2.6294_a0.3	2.629	1.454	0.3	0.056	−68 %	4.3	6.46	3.85	+22 %	6 of 9	1
7	G25_kp3.155_a0.3	3.155	1.454	0.3	0.065	−63 %	4.7	6.87	3.67	+16 %	7 of 9	1
8	G33_kp2.6294_a0.4	2.629	1.454	0.4	0.066	−63 %	4.6	6.90	2.98	−6 %	6 of 9	1

Repeat-to-repeat range of PWM jitter is about 0.012 % duty: settings closer together than that cannot be told apart. Amber row = on the EDGE of the acceptable region (fewer than 3/4 of the surrounding grid cells are acceptable): a small change in load could push it past the limit. Confirm the top few with more repeats.